Ethical and regulatory audit: NkwaPay credit scoring

ICS553 Prosit 2, MLE Group 1. Sources: docs/model_card.md, docs/datasheet.md, docs/compliance.md, learnings/11_fairness.md, ADRs 0019–0022.

Part 1. Disaggregated results: future Kenyan loans, step 7 recipe with all 13 clues, 2.1% cut-off, before the fixes

Attribute	Group	loans	defaults	approval rate	wrongly declined (good payers)	defaulters caught	F1	predicted default	actual default	PR-AUC	PR-AUC low	PR-A high
tenure	earlier loans, known outcomes	50513	644	0.917	0.079	0.466	0.123	0.010	0.013	0.075	0.061	0.102
tenure	earlier loans, no known outcome	1436	69	0.846	0.131	0.609	0.290	0.024	0.048	0.309	0.181	0.445
tenure	first loan	2514	275	0.447	0.530	0.738	0.244	0.074	0.109	0.274	0.227	0.328
size	Q1 smallest	13866	190	0.928	0.070	0.211	0.067	0.007	0.014	0.034	0.028	0.047
size	Q2	14182	224	0.903	0.088	0.629	0.177	0.009	0.016	0.114	0.095	0.148
size	Q3	13374	182	0.901	0.093	0.516	0.125	0.009	0.014	0.068	0.053	0.097
size	Q4 largest	13041	392	0.836	0.148	0.689	0.213	0.028	0.030	0.200	0.160	0.254
type	Type_1	49000	452	0.972	0.027	0.111	0.055	0.005	0.009	0.024	0.020	0.029
type	Type_10	400	10	0.722	0.269	0.600	0.099	0.026	0.025	0.424	0.006	0.705
type	Type_4	989	112	0.004	0.999	0.973	0.199	0.214	0.113	0.185	0.141	0.265
type	Type_5	800	56	0.046	0.953	0.964	0.132	0.061	0.070	0.072	0.048	0.115
type	Type_6	326	27	0.000	1.000	1.000	0.153	0.126	0.083	0.116	0.075	0.211
type	Type_7	2654	235	0.249	0.740	0.868	0.183	0.038	0.089	0.147	0.113	0.194
type	Type_9	190	36	0.000	1.000	1.000	0.319	0.195	0.189	0.235	0.179	0.349

Attribute	Group	loans	defaults	approval rate	wrongly declined (good payers)	defaulters caught	F1	predicted default	actual default	PR-AUC	PR-AUC low	PR-A high
type	other types	104	60	0.019	0.977	0.983	0.728	0.465	0.577	0.674	0.572	0.809
lender	Lender A	51654	699	0.932	0.063	0.395	0.131	0.009	0.014	0.083	0.068	0.100
lender	Lender B	2563	215	0.191	0.800	0.907	0.170	0.077	0.084	0.120	0.098	0.154
lender	Lender C	246	74	0.000	1.000	1.000	0.463	0.320	0.301	0.512	0.418	0.637
period	2022	50440	587	0.948	0.049	0.310	0.113	0.008	0.012	0.102	0.079	0.139
period	2023	1239	143	0.147	0.840	0.951	0.227	0.153	0.115	0.209	0.161	0.279
period	2024	2784	258	0.233	0.755	0.880	0.190	0.051	0.093	0.145	0.116	0.190
all	all future loans	54463	988	0.893	0.099	0.552	0.160	0.013	0.018	0.128	0.110	0.154

Ghana has no labels for us, so its selection rates and group default rates are in Part 3. Groups with few defaults have wide ranges.

Part 2. Proxy analysis: how well other clues reveal a group (ROC-AUC)

group	clues used	logistic	LightGBM
Ghana vs Kenya	all 13 clues	1.000	1.000
Ghana vs Kenya	without loan_type	0.991	0.994
Ghana vs Kenya	without loan_type and outcome clues	0.900	0.943
first loan	all 13 clues	1.000	1.000
first loan	loan details only (no history)	0.614	0.616
first loan	loan_type and duration only	0.590	0.593
'New Loan' flag	without the flag	0.971	0.972
'New Loan' flag	history clues only	0.957	0.957

group	clues used	logistic	LightGBM
Lender A vs others	all 13 clues	0.984	0.984
Lender A vs others	without loan_type	0.913	0.950
Lender A vs others	loan_type only	0.983	0.983
largest quarter of loans	all 13 clues	0.920	0.927
largest quarter of loans	without the two amount clues	0.627	0.648
largest quarter of loans	the two amount clues only	0.914	0.917

Part 3. Ghana: group default rates from Zindi's scores, with a cautious range counting customers

group	loans	defaults	rate	range_customers_lo	range_customers_hi
everyone	18594	722	0.039		
ghana_all	3525	447	0.127	0.099	0.161
ghana_first_loans	662	205	0.310	0.269	0.354
ghana_earlier_loans	2863	242	0.085	0.055	0.128
ghana_type_3	3039	213	0.070	0.042	0.114
ghana_type_2	449	226	0.503	0.438	0.568
ghana_over_90_days	479	231	0.482	0.420	0.545
ghana_13_clues_flagged	1672	355	0.212		
ghana_no_type_flagged	997	286	0.287		
ghana_final_flagged	781	262	0.335		
type_3_first_loans	296	29	0.098	0.064	0.146
type_2_first_loans	333	168	0.505	0.438	0.571
ghana_keep_type_no_often_flagged	1043	281	0.269		
ghana_unseen_as_type_1_flagged	542	237	0.437		

group	loans	defaults	rate	range_customers_lo	range_customers_hi
type_3_final_flagged	296	29	0.098		
type_3_13_clues_flagged	1186	121	0.102		
type_3_risk_band_1_of_5	609	31	0.051	0.018	0.134
type_3_risk_band_2_of_5	607	27	0.044	0.017	0.112
type_3_risk_band_3_of_5	607	30	0.049	0.021	0.110
type_3_risk_band_4_of_5	608	63	0.104	0.062	0.168
KENYA (everyone minus Ghana)	15069	275	0.018		

Part 4. Mitigations tested, Kenya

option	Kenya future-loan lift	cost per loan (KES)	cost with repeat business (KES)	full approval	any loan	first loans: good payers refused	other loans: good payers refused
no fix	7.779	47.930	86.730	0.893	0.893	0.530	0.080
first loans: starter loan, 10% of a typical loan	7.779	50.135	80.217	0.893	0.919	0.000	0.080
first loans: starter loan, 25% of a typical loan	7.779	52.324	82.406	0.893	0.919	0.000	0.080
first loans: starter loan, 50% of a typical loan	7.779	55.972	86.055	0.893	0.919	0.000	0.080
first loans: own cut-off, equal wrongly declined	7.779	52.265	87.694	0.903	0.903	0.325	0.080
first loans: own cut-off, equal approval	7.779	52.674	87.325	0.905	0.905	0.278	0.080
drop the 'borrow often' clues	7.715	47.342	85.173	0.895	0.895	0.506	0.078
hide tenure (no New_versus_Repeat, n_prev_loans, no_known_history)	8.174	47.748	88.583	0.888	0.888	0.573	0.083
reweigh tenure and outcome,	7.137	51.955	100.296	0.869	0.869	0.300	0.115

option	Kenya future-loan lift	cost per loan (KES)	cost with repeat business (KES)	full approval	any loan	first loans: good payers refused	other loans: good payers refused
honest Platt							
reweigh tenure and outcome, weighted Platt	7.137	52.303	102.385	0.865	0.865	0.300	0.120
drop loan_type (ADR 0019's Ghana model)	6.893	51.789	106.373	0.854	0.854	0.631	0.117
CHOSEN: drop 'borrow often' + starter loan at 10%	7.715	49.509	79.012	0.895	0.920	0.000	0.078

Part 5. Model Card

Model Card: NkwaPay default model

Following Mitchell et al. (2019). The evidence is in `notebooks/04_transfer_and_fairness.ipynb` Part 2 and `learnings/tables/11_*.csv`. This is a student project, not a licensed product.

1. Model details

- **Built by:** MLE Group 1 (Alice, Germain, Joseph, Liza), ICS553, Ashesi University, for the fictional lender NkwaPay.
- **Date and version:** November 2026, v1.
- **Type:** logistic regression with Platt calibration, fitted on out-of-fold scores from 5 customer-grouped folds (ADR 0016, 0017).
- **Clues:**
- **Kenya (10):** `loan_type`, `duration`, `New_versus_Repeat`, `n_prev_loans`, `n_finished`, `n_defaults`, `past_default_rate`, `no_known_history`, `mean_prev_amount`, `last_amount`.
- **Ghana:** the model doesn't decide anything yet. A rule by loan type does, using group default rates measured from Zindi's scores (ADR 0021).
- **Fairness choices:** none inside the training. Three "borrow often" clues were removed, and first-time applicants above the cut-off get a starter loan (ADR 0020).
- **Companion model:** LightGBM on log loan amount, for limits (ADR 0018). Not covered in the tables below.
- **Licence and contact:** course work; see the repository.

2. Intended use

- **Primary use:** estimate the chance that a short-term mobile-money loan defaults, to support an approve / starter loan / decline decision with up to 3 plain reasons.
- **Primary users:** NkwaPay's credit team, and borrowers through the reasons they receive.
- **Out of scope:**
- automatic decisions in Ghana before about 5,000 labelled Ghanaian loans exist;
- other countries, products, or business lending;
- setting prices;

- affordability checks: there is no income data;
- detecting loans from other lenders: there is no bureau data.

3. Factors

Factor	Relevant?	Evaluated?
Country	Yes	Partly: Ghana has no outcomes, so approval rates only
Tenure (first loan / earlier, no known outcome / known outcomes)	Yes	Yes
Loan type	Yes	Yes
Lender	Yes	Yes
Loan size	Yes	Yes
Time period	Yes	Yes
Gender	**Yes**	**No: not in the data.** Findex gives population context only
Age, income, region	Yes	No: not in the data

4. Metrics

- **Measures:** PR-AUC and lift (suited to about 2% defaults), ROC-AUC, F1, approval rate, good payers wrongly declined, defaulters caught, PPV and NPV, predicted vs actual default, cost per loan.
- **Cut-offs:** decline above 2.1% in Kenya and 3.2% in Ghana. These come from the cost of each mistake: 4,700 vs 100 KES, and 2,716 vs 90 GHS (ADR 0007).
- **Uncertainty:** customer-bootstrap 95% ranges.
- **Fairness priority:** sufficiency (an honest score) first, then good payers wrongly declined. Approval rates are reported but not forced equal, because groups' real default rates differ.

5. Evaluation data

- **Future Kenyan loans:** 54,463 loans, 988 defaults, 5,718 customers. Learn from older loans, score the next block.
- **Zindi test applicants:** 15,069 Kenyan and 3,525 Ghanaian loans; decisions only, as outcomes aren't available to us.
- **Preparation:** shared code in `src/features.py`; every step is fitted inside the training part.

6. Training data

68,464 labelled Kenyan loans, 6,508 customers, 2021–2024; 1.8% defaulted. See the datasheet.

Group	Loans	Default rate
First loan	6,363	6.7%
Earlier loans, no known outcome	5,567	1.7%
Earlier loans, known outcomes	56,534	1.3%
Type_1	61,723	0.8%
Other types	6,741	10.9%
Lender A	64,568	1.2%
Other lenders	3,896	12.2%

7. Quantitative analyses

Each factor alone, future loans at the 2.1% cut-off. These are for the 13-clue model, the baseline we audited. The chosen 10-clue model is within about 3 points on the tenure, size and Lender A rows; **Lender B gets worse**, from 80% to 87% wrongly declined (`11_fairness_after.csv`).

Group	Approved	Wrongly declined (95% range)	Defaulters caught	PR-AUC	Lift	Predicted / actual
All	89%	10% (9–11)	55%	0.13	7.1	1.3% / 1.8%
Known outcomes	92%	8% (7–9)	47%	0.08	5.9	1.0% / 1.3%
No known outcome	85%	13% (11–15)	61%	0.31	6.4	2.4% / 4.8%
First loan	45%	53% (51–56)	74%	0.27	2.5	7.4% / 10.9%
Smallest quarter	93%	7% (6–8)	21%	0.03	2.5	0.7% / 1.4%
Largest quarter	84%	15% (13–17)	69%	0.20	6.7	2.8% / 3.0%
Type_1	97%	3% (3–3)	11%	0.02	2.6	0.5% / 0.9%
Type_7	25%	74% (69–79)	87%	0.15	1.7	3.8% / 8.9%
Lender A	93%	6% (6–7)	40%	0.08	6.2	0.9% / 1.4%
Lender B	19%	80% (77–84)	91%	0.12	1.4	7.7% / 8.4%
2022	95%	5% (4–5)	31%	0.10	8.8	0.8% / 1.2%

Group	Approved	Wrongly declined (95% range)	Defaulters caught	PR-AUC	Lift	Predicted / actual
2023	15%	84% (78–91)	95%	0.21	1.8	15.3% / 11.5%
2024	23%	76% (71–80)	88%	0.15	1.6	5.1% / 9.3%

Where factors overlap (full table: `11_fairness_intersections.csv`):

Group	Loans	Defaults	Approved	Wrongly declined
First loan, largest quarter	456	92	15%	82%
First loan, smallest quarter	872	67	74%	27%
First loan, Lender A	1,883	163	60%	39%
First loan, Lender B	476	57	0.2%	99.8%
Known outcomes, Lender B	1,959	140	23%	77%

After the fixes (ADR 0020), same future loans:

	Before	After
Cost per loan, main costs	47.9 KES	48.9 KES
Cost per loan, at the measured repeat value (200 KES)	67.3 KES	64.0 KES
Lift	7.78	7.72
First-loan good payers refused	53%	4% (starter loan below 24% risk)
Other good payers refused	8%	8%
Declined applicants told to borrow more often (Zindi test, Kenya)	31%	0%

Applicants never seen by the model:

	Full loan	Starter or review	Declined
Kenya	82%	9%	9% (3% of first-

	Full loan	Starter or review	Declined
			timers)
Ghana (ADR 0021)	0%	86%	14%

8. Ethical considerations

- **Sensitive data:** financial records. There are no names or ID numbers beyond anonymous customer IDs. Nothing at row level leaves `data/`.
- **Effect on people:** a decision affects access to credit, and a loan can cause over-indebtedness.
- **Harms found and what we did:**
- first-time borrowers shut out completely → starter loan;
- the model rewarded frequent borrowing → clues removed.
- **Harms still open:**
- a starter loan can still push someone into debt;
- the limit model offers small borrowers up to 2.8× their usual loan;
- in Ghana, Type_3 borrowers aren't screened individually, and half of first-time applicants are refused because they ask for Type_2;
- the Ghana pilot loses about 10,000 GHS per 1,000 applicants unless good customers come back;
- the data covers only people lenders chose to lend to.
- **Risky uses:** using this model to decline Ghanaians at scale, or to chase repeat borrowing.

9. Caveats and recommendations

- **Barely validated on Ghana.** Group-level evidence from Zindi's scores puts Ghana's default rate at about 13% and Type_3's at 7%, above the 3.2% break-even (learnings/11 §10). The Ghana rule (ADR 0021) passes 4 of 6 stress tests: it's the best of 729 policies tested, not proven. Trust gates are in ADR 0019: about 5,000 labelled loans, recalibration, monthly drift checks, and a capped pilot.
- **Scores drift.** Default rates moved from about 1% (2022) to 9–12% (2023–24). Recalibrate often.
- **The starter loan is costed, not tested.** Its value rests on repeat business measured at about 200 KES, from Kenyan lending with no limit model. It breaks even at about 50–130 KES. Run it as a controlled pilot, and track outcomes.
- **Repeat the audit with gender** if NkwaPay collects it for bureau reporting.
- **The law asks for things this model can't do alone:** a bureau check, an affordability view, and appeal and record-keeping systems. See the compliance map.

Part 6. Datasheet

Datasheet: the data behind the NkwaPay models

Following Gebru et al. (2021). We didn't create these datasets, so many answers are "unknown". Those unknowns are findings in their own right. Counts come from `docs/01_data.md`.

Part 1: Zindi African Credit Scoring Challenge

Motivation

- **Why was it made?** For a machine-learning competition on predicting loan default. We use it as NkwaPay's stand-in loan book.
- **Who made and funded it?** Hosted by Zindi. The competition terms call the data "private property of the client". The client's identity is **unknown to us**.

Composition

- **What is one row?** One loan and lender pair. A loan shared between lenders has several rows (920 loan IDs appear in both files).
- **How many?** Train: 68,654 rows, 1,258 defaults (1.83%), Kenya only. After removing 190 loans not yet due: 68,464 loans, 6,508 customers. Test: 18,594 rows, no labels: 15,069 Kenyan and 3,525 Ghanaian (446 customers).
- **Representative? Unknown, and probably not.** Sampling isn't described. 92% of loans were sent in Jul–Dec 2022, and it holds only people lenders chose to lend to.
- **Label:** `target`, 1 = defaulted. The exact definition (days late, recoveries) is **not documented**.
- **Missing:** no blank cells in the raw files. In practice, **no Ghanaian has a known repayment outcome**.
- **Known errors or traps:**
- `Total_Amount_to_Repay` and `Lender_portion_to_be_repaid` include post-default penalties, so they leak the answer (PR-AUC 0.20 → 0.91);
- currency isn't stated: we assume local currency;
- the brief's `ProductCategory` and `IsDefaulted` don't exist in the data.
- **Subpopulations:** country, loan type, lender, new or repeat, loan size. **No gender, age, income or location.**
- **Could people be re-identified?** Customer IDs are anonymous, but repeated dates and amounts make a loan history fairly distinctive. So we treat rows as personal financial data.
- **Sensitive?** Yes, financial data.

Collection

- **How, when, by whom?** Loans sent Oct 2021 – Nov 2024. Method and collector **unknown**.
- **Consent and notice? Unknown.** We don't know whether borrowers were told or agreed to this use, or could withdraw. Ghana's Act 843 s.18(2) requires data about foreign people to be handled under their own law, which we can't confirm.
- **Ethical review or impact assessment? Unknown.** The BoG Directive requires a DPIA for a licensed lender (para. 2.1(5)(d)).

Preprocessing

- **What we did:** a feature audit (`learnings/03`); removed leaky columns and 190 not-yet-due loans; built borrower history counting a past loan only after its due date (`src/features.py`).
- **Raw data kept?** Yes, in `data/`, which is never committed.
- **Code:** public, in `src/` and `notebooks/`.

Uses

- **Used for:** this project's default and amount models, and the fairness audit.
- **Could cause unfair treatment:** first-time borrowers and small products are thin in the data, and Kenyan patterns may not apply elsewhere (ADR 0019).
- **Should not be used for:** scoring real borrowers without new consent and validation; pricing; identifying

anyone; judging groups it doesn't record, like gender.

Distribution

- Zindi account and competition rules. We redistribute nothing: the repository is public, and only aggregates are published. Raw rows aren't uploaded to AI tools.

Maintenance

- We don't maintain the source. No known errata process. The competition closed on 12 Jan 2025; late scoring still works. Local copies should be deleted when the course ends.

Part 2: Global Findex Ghana 2021

- **Motivation:** made by the World Bank to measure financial inclusion. We use it to describe who lacks credit in Ghana (Part D4).
- **Composition:** one row per surveyed adult; 1,000 people, 120 columns, with weights (`wgt`). Includes gender, age, education, income quintile and rural/urban. **No region.**
- **Collection:** a national survey, 2021. Weighted shares match published figures (68% any account, 60% mobile money, 74.2% of men with an account).
- **Preprocessing:** none beyond recoding answers. All results are weighted.
- **Uses:** population context only. **It can't be linked to borrowers** and doesn't test our model.
- **Distribution:** World Bank Microdata Library; registration required. We publish aggregates only.
- **Maintenance:** by the World Bank. Rural/urban and education codes should be confirmed with the codebook, which we haven't downloaded.

Part 7. Regulatory map and borrower letters

Compliance map and borrower letter

What the design meets and doesn't meet under Ghanaian law. Clause detail and sources are in `knowledge/regulation.md`. We are students, so this isn't legal advice. We did not read Act 1052, L.I. 2394 or BoG's 2017 disclosure rules, which may add requirements.

Worth noting: the BoG Directive has no rule on algorithms, fairness or automated decisions. Fairness here rests on its consumer-protection goals and on Act 843.

What the design meets

Requirement	Source	How
Explain the logic of an automated decision	Act 843 s.35(1)(d)	Up to 3 plain reasons per decision; they add up exactly to the model's score (ADR 0017)
Clear, accurate, not misleading information	Directive 6(1)(a), (d)	Reasons use the applicant's own numbers; every sentence is tested in `src/reasons.py`
No contacts or social-media	Directive 7.2(c)–(d)	Not used

Requirement	Source	How
data		
No race, ethnicity, religion or politics	Act 843 s.37; Act 726 s.24(4)	Not in the data or the clues
Don't penalise new borrowers unfairly	Act 726 s.21(1)(p), in spirit	Starter loan rather than a decline (ADR 0020)
Responsible credit	Directive 1.3	"Borrow often" clues removed; capped first loans
Keep personal data secure	Act 843 s.28	No rows in the public repository; aggregates only

What the design doesn't meet yet

#	Requirement	Source	Gap	What would fix it
1	**Search a credit bureau before every decision**	Act 726 s.26(4)	**The clearest gap.** No bureau feed; only the lender's own records	A bureau check before scoring, with consent (s.26(1); Directive 7.3)
2	Handle foreign data under its own law	Act 843 s.18(2)	Consent for the Kenyan data is unknown	Retrain on NkwaPay's own consented data
3	Register with the DPC; an approved DPIA	Act 843 s.27, s.46; Directive 2.1(5)(d)	Organisational; can't be done by a student project	The Model Card and datasheet feed the DPIA
4	Review route; complaints resolved in 20 days	Act 843 s.41; Directive 6(1)(e)	The scorer has no appeal path	A digital "ask for review" step and a log
5	**Review must be digital**	Directive 2.3(1)	ADR 0019 says "manual review"; the physical office can't do it	Use a starter loan or digital review only
6	Keep decision records	Act 843 s.24, s.35; Act 726 s.35	The scorer returns a result but stores nothing	An append-only log of inputs, version, score, cut-off, reasons, consent
7	Single-customer limit	Directive 3.5(5)	BoG's figure isn't published in the Directive; the limit has no cap	A configurable cap, once known
8	Avoid over-indebtedness and stacking	Directive 1.3(3), 7.1(2); Act 1052 unverified	Can't see other lenders' loans; no income data	Bureau data; affordability check
9	Valid for Ghanaian borrowers	Directive 3.5(1),	Never tested on Ghanaian	Capped pilot (ADR 0021);

#	Requirement	Source	Gap	What would fix it
		4(2)	outcomes	the ADR 0019 trust gates
10	Total-cost and no-compounding disclosure	Directive 6(1)(b)–(c), 7.1(3)	Outside the model	The app shows amount, due date and total repayable
11	Gender fairness (asked by the brief, not found in these laws)	Brief Part D	No gender in the data	Audit it if collected (Act 726 s.24(2)(f))

Letter to a declined borrower

The reasons come from `src/reasons.py`. This applicant is made up.

> **We could not approve this loan today.**
>
> This was an automated decision. The main reasons were: 1. This kind of loan (Type_7) has been repaid less often than others. 2. The loan is for 30 days; longer loans are repaid less often. 3. Your most recent loan (9,000) was larger than your usual loan (3,000).
>
> **What you can do** - You can ask us to explain or review this decision. Reply HELP. We will answer within 20 days. - You can ask to see the information we hold about you, and to correct it. - You may complain to the Bank of Ghana.
>
> This decision doesn't stop you from applying again.

And for a first-time applicant, who gets a smaller loan rather than a no:

> **We can offer you a starter loan of GHS 300 today.**
>
> This was an automated decision. This is your first loan with us, so we have no borrowing history to go on. Repaying on time helps us offer you more.
>
> Before you accept, we will show you the amount, the due date and the total you must repay. You can ask us to explain or review this decision by replying HELP, and we will answer within 20 days.

And for a Ghanaian applicant asking for a product we aren't offering yet (ADR 0021):

> **We can't offer this kind of loan (Type_2) in Ghana yet, but you can ask for a starter loan of GHS 300 instead.**
>
> This was an automated decision. It is about the product, not about you: about half of loans of this kind have not been repaid in Ghana.
>
> **What you can do** - Ask for a Type_3 starter loan. We will show you the amount, the due date and the total you must repay before you accept. - Ask us to explain or review this decision. Reply HELP, and we will answer within 20 days. - Ask to see the information we hold about you, and to correct it. You may complain to the Bank of Ghana.

Why the first-time letter is a loan, not a no: our notes warn against "you are a new customer" as the only reason for a decline (Act 726 s.21(1)(p), in spirit). Without the starter loan, that is what first-timers would get. With it, the sentence explains a smaller amount, not a refusal.

Wording rules: no blame; no protected traits; never tell someone to borrow more to qualify; never reveal the model's weights (Act 843 s.35(2) allows this).